Generational Depth Estimation of Recursive Synthetic Text: Measuring the Discrimination Boundary

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Abstract

Web-scale corpora are increasingly contaminated by recursive synthetic content, yet no method estimates a document's generational depth. We measure a lower bound on the generational discrimination boundary, $K^*_{\text{empirical}}$ (the maximum depth at which adjacent generations remain distinguishable). Under prompt-chain generation across 10 model configurations (6 architecture families, 1B–8B), 4 source domains, and 4 decoding strategies, $K^*_{\text{empirical}} \leq 2$ for all web text configurations (discrimination threshold θ=0.60); structured academic text (arXiv) reaches $K^*_{\text{empirical}}$=3 across four independent architecture families. Beyond two generation passes, web text becomes statistically indistinguishable across deeper generations, a *one-wash laundering* effect that concentrates the dominant exploitable signal at the human-vs.-machine boundary. Across continuation-mode generators, generator-agnostic statistical features enable 80–92% cross-model transfer with scorer-invariant feature rankings ($\rho > 0.90$); depth estimation on corpora from unknown generators requires no retraining. Under extreme contamination (∼83% synthetic), $K^*_{\text{empirical}}$ correctly predicts that binary filtering outperforms graduated alternatives; at realistic ratios (≤50%), strategy differences remain below 0.3 pp[1].

1 Introduction

Web-scale text corpora are increasingly contaminated by synthetic content that has itself been generated from earlier synthetic content. Estimates suggest that a substantial fraction of online text is machine-generated (Sun et al., 2025; Spennemann, 2025), and this share continues to rise. When models train on the outputs of earlier models, the resulting multi-generational synthetic text degrades with each iteration: tail modes vanish, lexical diversity contracts, and surprisal distributions flatten (Shumailov et al., 2024; Dohmatob et al., 2024). This degradation arises in both *retrain-chains* (iteratively training on model outputs) and *prompt-chains* (recursively generating from previous outputs without retraining); we study the latter, which isolates generation-level information loss from training dynamics (Appendix B.7). Yet the dominant paradigm treats all machine-generated text identically, discarding the generational depth dimension.

This gap cuts across four lines of work: binary detectors (Basani and Chen, 2026; Zhang et al., 2025) produce no depth estimate (Section 4.6); membership inference (Zawalski et al., 2026) addresses a different question; the model collapse literature (Shumailov et al., 2024; Alemohammad et al., 2024; Schaeffer et al., 2025) provides no per-text tool; and data selection methods (Xie et al., 2023b; Tirumala et al., 2023; Drayson et al., 2025) ignore recursive depth. No existing method estimates how many rounds of recursive generation a text has undergone.

The Data Processing Inequality (DPI) guarantees that information degrades monotonically across generations, motivating a finite discrimination boundary K^*. We empirically measure a lower bound $K^*_{\text{empirical}}$ on the discrimination boundary (the maximum depth at which a classifier reliably distinguishes adjacent generations) using 15 statistical features and an Ordinal Binary Decomposition (OBD) LightGBM classifier. We operationalize $K^*_{\text{empirical}}$ as the deepest $k \geq 1$ where the pairwise area under the ROC curve ($\text{AUC}_{k-1,k}$) exceeds $\theta = 0.60$ (Section 3.1).

Our contributions:

(a) **Measuring the discrimination boundary**: $K^*_{\text{empirical}} \leq 2$ across all 10 model configurations (6 architecture families, 1B–8B) and 4 decoding strategies; $K^*_{\text{empirical}}$=2 for 6/10

[1]Code and data are available at https://anonymous.4open.science/r/gen-depth-contamination-E6C1/.

spanning three families ($p < 0.001$; Section 4.1). Beyond two passes, web text becomes statistically indistinguishable; both supervised OBD and zero-shot LLMs fail beyond $K^*_{\text{empirical}}$ (McNemar $p > 0.05$; Section 6).

(b) **Domain dependence**: C4, Wikipedia, and code yield $K^*_{\text{empirical}} \leq 2$; structured academic text (arXiv) yields $K^*_{\text{empirical}} = 3$ across four families (Section 4.5).

(c) **Generator-agnostic transfer** (continuation mode): 15 statistical features enable 80–92% accuracy retention with scorer-invariant rankings ($\rho > 0.90$); $K^*_{\text{empirical}}$ agreement holds across scorers for strong-signal configurations (Section 4.2).

As a consistency check under extreme contamination ($\sim$83% synthetic), $K^*_{\text{empirical}} = 2$ correctly predicts the strategy ordering: binary curation minimizes Massive Multitask Language Understanding (MMLU) degradation (0.5941 vs. base 0.6018); at realistic ratios ($\leq$50%), all strategies converge (Section 5).

2 Related Work

Model collapse and recursive degradation. Iterative training on model outputs causes tail-distribution degradation (Shumailov et al., 2024; Alemohammad et al., 2024; Dohmatob et al., 2024, 2025), diversity contraction (Guo et al., 2024; Padmakumar and He, 2024; Briesch et al., 2023), and predictable convergence dynamics (Schaeffer et al., 2025; Kazdan et al., 2025; Gerstgrasser et al., 2024; Wang et al., 2025). Fu et al. (Fu et al., 2025) formalize the convergence via Markov chain theory, complementing the DPI's mutual information monotonicity. Our prompt-chain operates in the replace regime (Kazdan et al., 2025), making $K^*_{\text{empirical}} = 2$ a conservative lower bound; empirically, detection accuracy erodes with each generation (Hao et al., 2025), consistent with our measured discrimination boundary. This literature characterizes macro dynamics but provides no per-text depth estimation, the gap we address.

Machine-generated text detection. Statistical detectors (Mitchell et al., 2023; Bao et al., 2024) and model-based classifiers (Hans et al., 2024; Basani and Chen, 2026; Zhang et al., 2025; Verma et al., 2024) achieve strong human-vs.-machine separation; benchmarks such as RAID (Dugan et al., 2024) evaluate robustness at scale. Recent extensions move beyond binary labels: Cheng et al. (Cheng et al., 2025) estimate human-AI collaboration degree, Mehta et al. (Mehta, 2025) propose contamination hierarchies, and Wang et al. (Wang et al., 2024) construct multigenerator benchmarks. Robustness concerns arise under paraphrasing (Sadasivan et al., 2025; Krishna et al., 2023), cross-mode transfer (Hao et al., 2025), and demographic bias (Liang et al., 2023); CoDeC (Zawalski et al., 2026) addresses contamination via membership inference. None of these methods estimates recursive generation depth.

Data curation, ordinal methods, and watermarking. Data Selection with Importance Resampling (DSIR) (Xie et al., 2023b), D4 (Tirumala et al., 2023), and DoReMi (Xie et al., 2023a) select training data by domain relevance but are depth-agnostic; Drayson et al. (Drayson et al., 2025) combined binary detection with importance resampling, a special case when $K^* = 1$. Our OBD classifier follows Frank & Hall's (Frank and Hall, 2001) binary decomposition for ordinal regression; our neural probe extends probing (Alain and Bengio, 2017; Belinkov, 2022) to the discrimination boundary. Text watermarking (Kirchenbauer et al., 2023) embeds provenance signals proactively; our work addresses the complementary post-hoc scenario where the generator is not controlled.

3 Method

3.1 Problem Formulation

A generation operator G_δ (LLM + decoding strategy δ) produces successive generations $X \to Y_1 \to \cdots \to Y_K$, where $Y_k = G_\delta(Y_{k-1})$ and $Y_0 \triangleq X$ (Figure 1). This sequence forms a Markov chain. By the Data Processing Inequality, $I(X;Y_1) \geq I(X;Y_2) \geq \cdots \geq 0$: mutual information with the seed text degrades monotonically. Together with Markov chain convergence, this motivates the existence of a finite discrimination boundary where $\text{AUC}_{k-1,k}$ falls below a decision threshold, though AUC and mutual information are not formally equivalent; the actual saturation depth must therefore be measured empirically. Pairwise AUC generally decreases with depth but may exhibit noise-level reversals (e.g., Pythia-1.4B 3v4 AUC $= 0.505$ vs. 4v5 $= 0.508$, a difference within bootstrap CI; Table 10), reinforcing that $K^*_{\text{empirical}}$ is a measured quantity rather than a theoretical prediction.

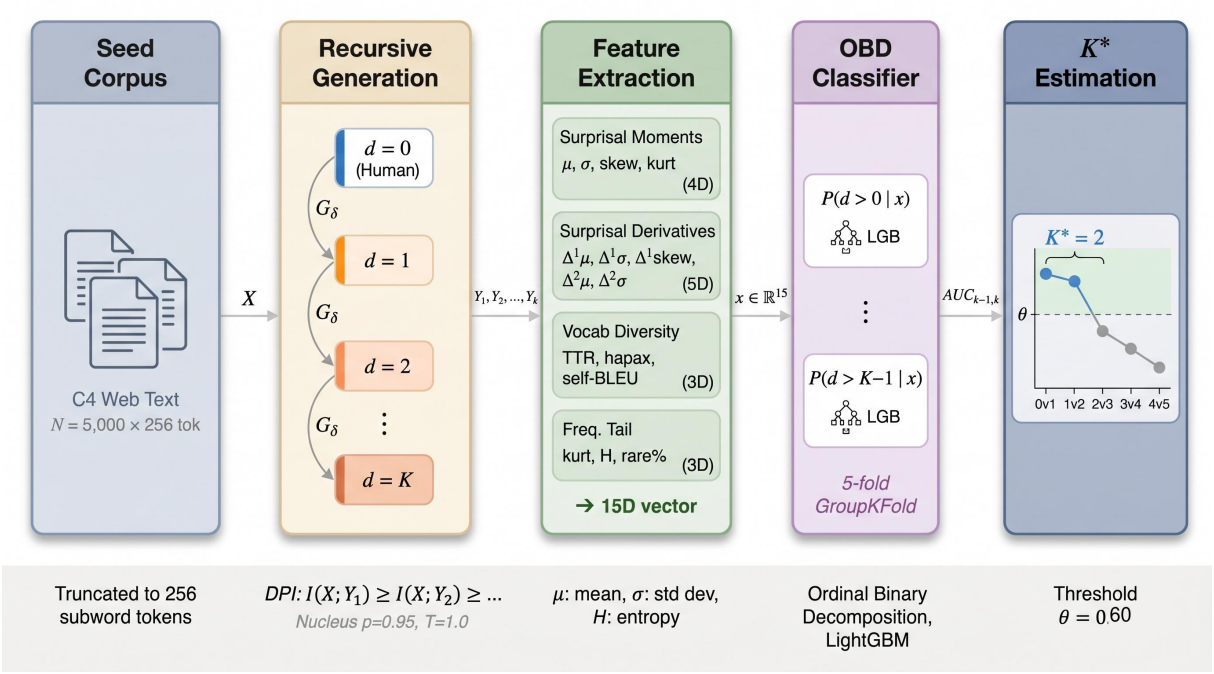


Figure 1: Generational depth estimation pipeline. Seed documents are recursively passed through a generation operator G_δ to produce a depth chain. Corpus-level statistical features are fed to the OBD-LightGBM classifier for depth prediction. (Simplified notation; see Eq. 1 for the formal $K^*_{\text{empirical}}$ definition.)

We define the operational lower bound:

$$K^*_{\text{empirical}} = \max\{k \geq 1 : \text{AUC}_{j-1,j} > \theta,\ \forall 1 \leq j \leq k\}, \quad (1)$$

which requires all pairs up to depth k to exceed the threshold (the $\forall j$ constraint is naturally satisfied because $\text{AUC}_{k-1,k}$ decreases monotonically above θ across all tested configurations; Table 2). The theoretical upper bound K^*_{theory} (the true maximum discriminable depth given optimal features) is not directly estimated in this work. Since any specific classifier on a feature subset achieves AUC at most equal to the Bayes-optimal classifier on all features, $K^*_{\text{empirical}} \leq K^*_{\text{theory}}$ follows directly. The DPI motivates the empirical observation that $\text{AUC}_{k-1,k}$ generally decreases with k under non-degenerate decoding, though AUC is not a monotone function of MI (as noted above). Here $\text{AUC}_{k-1,k}$ is the point-estimate pairwise AUC and $\theta = 0.60$ is a reference operating point; θ sits between the 1v2 AUC cluster (0.607–0.696) and the 2v3 cluster (0.55–0.59), so that $K^*_{\text{empirical}}$ is insensitive to the exact threshold within this range (justified below; Figure 3). All pairwise significance tests use 10,000-iteration permutation tests with Bonferroni correction for 10 comparisons ($\alpha_{\text{corrected}}$=0.005); we report bootstrap 95% confidence intervals (CIs) (1,000 resamples unless otherwise noted) for all boundary-pair AUCs and explicitly mark marginal cases where Bonferroni-corrected lower bounds fall below θ (Section 4.1).

Threshold justification. θ=0.60 is a reference operating point supported by two independent lines of evidence. First, a 10,000-iteration permutation test yields a 95th-percentile null of <0.55 ($p < 0.002$ for all boundary comparisons). Second, $K^*_{\text{empirical}}$=2 is insensitive to the exact threshold: 1v2 AUC values cluster at 0.607–0.696 while 2v3 values cluster at ∼0.54–0.59, a ∼1–10 pp discontinuity that ensures $K^*_{\text{empirical}}$ is stable across $\theta \in [0.59, 0.60]$; at θ=0.61, one borderline case (Pythia-1.4B, 1v2 AUC=0.607) shifts to $K^*_{\text{empirical}}$=1. Only at extremes does $K^*_{\text{empirical}}$ shift broadly (θ=0.55: $K^*_{\text{empirical}}$=3 for two Qwen configurations; θ=0.70: $K^*_{\text{empirical}}$=1 across all). Figure 3 provides the full sensitivity surface across $\theta \in [0.50, 0.70]$. The downstream filtering evaluation (Section 5) uses the OBD classifier without re-optimizing θ, avoiding circularity.

3.2 Features and Classifier

Each document is characterized by a 15-dimensional feature vector spanning four groups: *surprisal moments* (mean, std, skewness, kurtosis; 4 dims), *surprisal derivatives* (mean and std of first- and second-order differences, plus first-order skewness; 5 dims), *vocabulary diversity* (type-token

Table 1: Experimental configurations for depth-chain generation (six primary configurations).

Config	Model	Mode	N
Qwen-rw	Qwen2.5-1.5B-Inst	Rewrite	5,000
Qwen-cont	Qwen2.5-1.5B-Inst	Cont.	5,000
Qwen-base	Qwen2.5-1.5B	Cont.	5,000
Pythia	Pythia-1.4B	Cont.	5,000
OLMo	OLMo-1B	Cont.	5,000
Qwen-7B	Qwen2.5-7B	Cont.	5,000

ratio (TTR), hapax legomena ratio, self-BLEU; 3 dims), and *frequency tail statistics* (frequency kurtosis, entropy, rare token fraction; 3 dims). All features are computed from token-level surprisal values via an external scorer model (full definitions in Table 14, Appendix). Self-BLEU reference sets are drawn exclusively within each fold to prevent information leakage across cross-validation splits; self-BLEU ranks #1 in 1/8 conditions (greedy decoding) but is absent from the top-3 in all 7 others. Removing self-BLEU reduces mean AUC_{1v2} by < 1 pp across all models (14D $K^*_{\text{empirical}} = 2$ for all 4 configurations); in deployment where reference corpora are unavailable, self-BLEU can be replaced with corpus-level self-similarity or omitted entirely, leaving 14 features (surprisal statistics, TTR, hapax ratio, frequency tail) that require no external reference and retain full $K^*_{\text{empirical}}$. These features are generator-agnostic in that they require no knowledge of or access to the generating model, though they depend on an external scorer for surprisal computation (Section 4.2).

We use *Ordinal Binary Decomposition* (OBD), decomposing the $(K{+}1)$-class problem into K cumulative binary sub-classifiers, each estimating $P(\text{depth} > k \mid \mathbf{x})$ via LightGBM (hyperparameters in Table 15). OBD yields superior pairwise AUC compared to ordinal logistic regression (mean 0.6099 vs. 0.4883 across three model families), whose low MAE stems from collapsing predictions toward central depths rather than improved discrimination (Appendix B.13).

3.3 Experimental Setup

We evaluate six primary configurations spanning three model families (Table 1): Qwen2.5-1.5B-Instruct (rewrite and continuation), Qwen2.5-1.5B (base), Qwen2.5-7B, Pythia-1.4B, and OLMo-1B; four additional models are introduced in Section 4. Each uses 5,000 C4 seeds truncated to 256 tokens, recursively generated for $K{=}5$ iterations (30,000 documents per configuration; Table 8). Default decoding: nucleus $p{=}0.95$, $T{=}1.0$; decoding ablation covers greedy, nucleus $p{=}0.9$, $T{=}0.7$, and $T{=}1.2$. Continuation mode truncates seeds to 128 tokens and extends to 256; rewrite mode paraphrases the full text. Evaluation: 5-fold GroupKFold cross-validation grouped by seed document; pairwise AUC as primary metric.

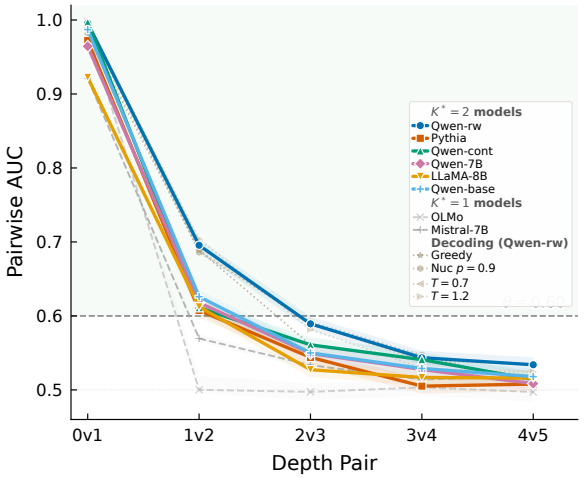


Figure 2: Pairwise AUC decay across adjacent depth pairs (0v1, 1v2, …, 4v5) for model-mode configurations (solid) and decoding strategies (dashed); shaded bands show bootstrap 95% CIs. Horizontal dashed line: $\theta = 0.60$.

4 Experiments

We evaluate on ten configurations spanning six families: Pythia-1.4B/6.9B, OLMo-1B, Qwen2.5-1.5B-Instruct (rewrite/continuation), Qwen2.5-1.5B (base), Qwen2.5-7B, LLaMA-3.1-8B, Mistral-7B, and Gemma-2-2B. Unless noted, all experiments use the OBD-LightGBM classifier with 15 features (Section 3), 5,000 C4 texts per depth level (depths 0–5).

4.1 The Discrimination Boundary

Table 2 and Figure 2 show $K^*_{\text{empirical}} \leq 2$ across all 10 configurations: 6/10 yield $K^*_{\text{empirical}}{=}2$ (1v2 AUC: 0.607–0.696; $p < 0.001$), while 4/10 yield $K^*_{\text{empirical}}{=}1$ (1v2 AUC: 0.500–0.598; Section 6). Across six architecture families, three (Qwen, Pythia, LLaMA) yield $K^*_{\text{empirical}}{\geq}2$ in at least one configuration; the remaining three (OLMo, Gemma-2, Mistral) yield $K^*_{\text{empirical}}{=}1$. Pythia exhibits a scale-dependent split (1.4B: $K^*_{\text{empirical}}{=}2$, 6.9B: $K^*_{\text{empirical}}{=}1$); Qwen dominates the $K^*_{\text{empirical}}{=}2$ count (4 of 6 configurations). Under Bonferroni correction (99.5% CIs;

Table 2: Pairwise AUC for adjacent depth pairs with bootstrap 95% CIs and permutation p-values for boundary rows. $K^*_{\text{empirical}} = 2$ requires 1v2 AUC $> \theta$=0.60 and 2v3 AUC $< \theta$; CIs quantify boundary uncertainty. Extended results (all models) in Table 10.

	Qwen-rw		Pythia		OLMo		Qwen-cont		Qwen-7B‡	
	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI
0v1	.9956		.9722		.9861		.9971		.9644	[.961, .967]
1v2	.6955	[.686, .704]*	.6074	[.598, .617]*	.5001	[.496, .519]	.6112	[.602, .621]*	.6178	[.608, .626]*
2v3	.5893	[.580, .599]	.5440	[.534, .554]	.4973	[.486, .509]	.5609	[.551, .571]	.5492	[.539, .559]
3v4	.5435		.5051		.5034§		.5409		.5276	
4v5	.5340		.5079		.4971		.5141		.5085	

*Permutation $p < 0.001$ for all starred boundary comparisons (AUC $> \theta$).
‡Generator: Qwen2.5-7B (fp16); scorer: Qwen2.5-1.5B.
§OLMo 2v3 (0.497) < 3v4 (0.503): non-monotonic reversal within noise ($\Delta = 0.006$).
Full results for all 10 configurations (including LLaMA-8B, Mistral-7B, Qwen-base, Gemma-2-2B, and Pythia-6.9B) in Table 10.

Table 27), three of six $K^*_{\text{empirical}}$=2 configurations retain significance; the three marginal cases (Pythia-1.4B, Qwen-cont, LLaMA-8B; CI lower bounds 0.593–0.599) reflect narrow effect sizes rather than classification failure ($K^*_{\text{empirical}}$ is robust to θ within $[0.59, 0.60]$; Figure 3). LLaMA-3.1-8B confirms $K^*_{\text{empirical}}$=2 (0v1 AUC $= 0.9225$, 1v2 AUC $= 0.6123$ $[0.603, 0.622]$; CI lower bound above θ; Table 10). All four decoding strategies likewise yield $K^*_{\text{empirical}}$=2 (Table 9; Pythia-1.4B replication in Table 19). The 1v2 spread due to generation mode ($\Delta = 8.4$ pp within Qwen) exceeds the spread across $K^*_{\text{empirical}}$=2 continuation models ($\Delta = 1.9$ pp). Bootstrap resampling (1,000 iterations) confirms stability: 8/10 configurations yield identical $K^*_{\text{empirical}}$ across all resamples (two boundary cases have CIs of width 1; the $K^*_{\text{empirical}} \leq 2$ bound holds universally); together with the between-sample 5K-vs.-10K comparison (Table 28), this bounds $K^*_{\text{empirical}}$ uncertainty from complementary angles (Table 29; per-fold AUCs in Table 17). The two exceptions, Gemma-2B C4 and LLaMA-8B Wikipedia (92.7% and 94.3% modal-$K^*_{\text{empirical}}$ consistency), have 1v2 AUC within 1 pp of θ; their bootstrap instability predicts the between-sample drift in Table 28; boundary proximity rather than sample size drives the observed variability.

Scale and instruction-tuning invariance. $K^*_{\text{empirical}}$=2 persists from 1.5B through 8B: Qwen-7B (1v2 AUC $= 0.6178$ $[0.608, 0.626]$) and LLaMA-3.1-8B (1v2 AUC $= 0.6123$) both exceed θ, despite progressively reduced 0v1 signal (0.9644 and 0.9225 vs. > 0.97 at $\leq$1.5B). A base-vs.-instruct ablation at 1.5B likewise yields $K^*_{\text{empirical}} = 2$ (1v2 AUC $= 0.626$); the boundary is not an instruction-tuning artifact.

Figure 3 displays the sensitivity surface across $\theta \in [0.50, 0.70]$: four of five configurations shown yield $K^*_{\text{empirical}} \geq 2$ for $\theta \leq 0.60$ (OLMo: $K^*_{\text{empirical}} = 1$ throughout); at θ=0.55, two Qwen configurations reach $K^*_{\text{empirical}} = 3$; at θ=0.70, all drop to $K^*_{\text{empirical}} = 1$. $K^*_{\text{empirical}}$=2 is stable across $\theta \in [0.59, 0.60]$; at θ=0.61, one borderline case (Pythia-1.4B) drops to $K^*_{\text{empirical}}$=1.

4.2 Cross-Model Transfer

Continuation-mode classifiers transfer well across generators (Figure 4; full matrix in Table 7): Pythia↔OLMo retains 90–92%, Qwen-cont↔Pythia 84–85%, Qwen-cont↔OLMo 80–85%. Cross-scale transfer from Gemma-2-27B retains >100% accuracy on 1B–8B models (101.3% avg; ≤1.5B→27B: 76.5%); larger models may learn a universal depth-feature superset (Table 31; filtering validation in Table 32). Qwen-rewrite classifiers transfer at 61–63% to Pythia and OLMo but retain 82% to Qwen-cont; cross-architecture degradation reflects a mode-specific artifact, while the within-family retention points to a Qwen-specific component layered on top.

Cross-scorer validation. Replacing the self-scorer with Pythia-1.4B or OLMo-1B preserves feature rankings ($\rho > 0.90$) but attenuates absolute discriminability at the boundary (Table 23; $K^*_{\text{empirical}}$ comparison in Table 22). Feature ranking independence is a strong result: the same features dominate under all scorers (Table 23). $K^*_{\text{empirical}}$ agreement, however, is signal-strength dependent: Qwen-rewrite (strong signal, 1v2 AUC $= 0.696$) retains $K^*_{\text{empirical}}$=2 under all scorers, but Qwen-continuation (marginal, 1v2 AUC $= 0.611$, only 0.011 above θ) drops to $K^*_{\text{empirical}}$=1 (Pythia:

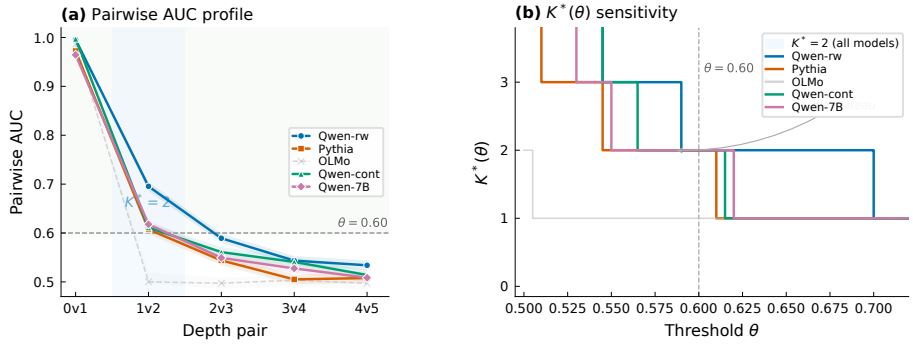


Figure 3: $K^*_{\text{empirical}}(\theta)$ sensitivity curve for five configurations. $K^*_{\text{empirical}}$: maximum discriminable depth; θ: AUC decision threshold. Four of five configurations yield $K^*_{\text{empirical}}$=2 at θ=0.60 (dashed line); OLMo yields $K^*_{\text{empirical}}$=1. At θ=0.70, all drop to $K^*_{\text{empirical}}$=1.

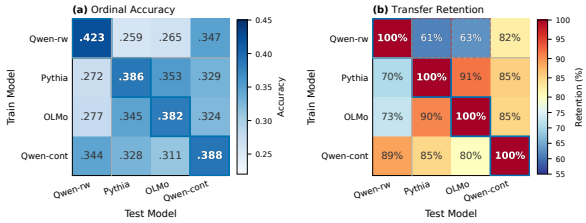


Figure 4: Cross-model transfer heatmap (three continuation-mode models plus Qwen-rewrite). **(a)** 6-class ordinal accuracy when trained on the row model and evaluated on the column model; diagonal entries (highlighted) are in-domain baselines. **(b)** Retention rate (off-diagonal accuracy divided by in-domain accuracy). Continuation-mode pairs retain 80–92%; Qwen-rw transfers at 61–63% to other architectures but 82% to Qwen-cont, consistent with a mode-specific artifact.

0.5923; OLMo: 0.5858). A three-way analysis of variance (ANOVA) (Generator × Scorer × Depth-Pair; Table 24) confirms a statistically significant but modest Scorer main effect (η^2_p=.192, $p < .001$) and a significant Generator×Scorer interaction (η^2_p=.170); absolute AUC values are thus scorer-dependent even when $K^*_{\text{empirical}}$ classifications largely agree.

4.3 Feature Ordering and Generation Mode

Among continuation-mode models, single-feature AUC rankings are highly correlated ($\rho \geq 0.85$; Table 6). The top-3 features (`surp_d1_std`, `surp_std`, `surp_d2_std`) are shared across all three families: autoregressive generation progressively damps surprisal variability. Qwen-rewrite diverges (`freq_entropy` dominates), consistent with its low cross-architecture transfer (61–63% to Pythia/OLMo; 82% within Qwen). Within Qwen, rewrite yields 1v2 AUC = 0.6955 vs. continuation's 0.6112 ($\Delta = +8.4$ pp); continuation mode is therefore recommended as the canonical condition.

Table 3: Neural probe pairwise AUC using frozen-encoder MLP. Each cell reports AUC for classifying depth d vs. d+1 text. $K^*_{\text{empirical}}{}^{\text{neural}}$: maximum depth where AUC exceeds $\theta = 0.60$.

Model	2v3 AUC	95% CI	$K^*_{\text{empirical}}{}^{\text{neural}}$
Qwen-1.5B	0.6159	[0.610, 0.622]	3
Pythia-1.4B	0.5963	[0.589, 0.603]	2
OLMo-1B	0.5056	[0.497, 0.516]	1†

†OLMo 1v2 AUC = 0.5054 (below θ); $K^*_{\text{empirical}}{}^{\text{neural}} = 1$.

4.4 Neural Probe

Replacing the 15D features with frozen encoder hidden states (multilayer perceptron (MLP) probe; Table 3; architecture in Table 16), Qwen achieves $K^*_{\text{empirical}}{}^{\text{neural}} = 3$ (2v3 AUC = 0.6159), while Pythia remains at $K^*_{\text{empirical}}{}^{\text{neural}} = 2$ ($\text{AUC}_{2v3} = 0.5963$); OLMo yields $K^*_{\text{empirical}}{}^{\text{neural}} = 1$ (1v2 AUC = 0.5054, consistent with its 15D boundary). We hypothesize that the Qwen-only elevation reflects self-familiarity artifacts, consistent with the poor transfer observed in Section 4.2; the 15D features yield $K^*_{\text{empirical}} = 2$ for most families (OLMo: $K^*_{\text{empirical}} = 1$).

4.5 Cross-Domain Variation

The preceding experiments use C4 (general web text). To test domain dependence, we evaluate on Wikipedia (n=5,000) and arXiv abstracts (n=2,200) across five architecture families (Table 21). Wikipedia yields $K^*_{\text{empirical}}$=2 for Qwen-1.5B (1v2 AUC = 0.619; Table 4), but $K^*_{\text{empirical}}$=1 for LLaMA-8B and Mistral-7B (1v2 AUC = 0.592, 0.558; both below

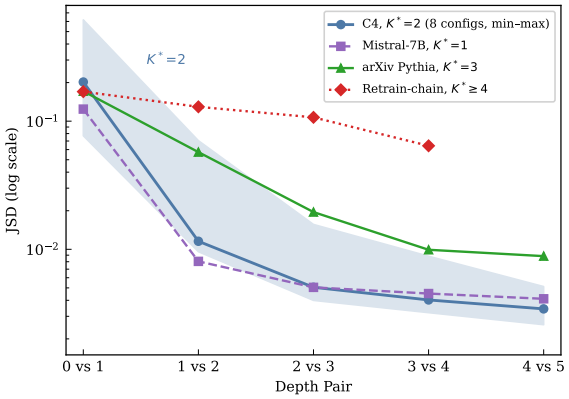


Figure 5: JSD between adjacent depth pairs (log scale). C4 band (blue): min-max over 8 $K^*_{\text{empirical}}$=2 configurations (4 model-mode pairs + 4 Pythia decoding variants; $K^*_{\text{empirical}}$=1 models excluded). arXiv (green) sustains signal to depth 3; retrain-chain (red) remains elevated.

θ). Qwen-1.5B, Pythia-1.4B, LLaMA-3.1-8B, and Mistral-7B yield $K^*_{\text{empirical}}$=3 for arXiv (Table 4): 2v3 AUC = 0.643 [0.628, 0.658], 0.655 [0.634, 0.662], 0.622 [0.605, 0.635], and 0.628 respectively (permutation $p < 0.001$); structured academic text thus permits discrimination to greater depth independently of architecture. OLMo-1B is the sole exception ($K^*_{\text{empirical}}$=1 across all domains; cross-scorer validated, p=0.510); Mistral-7B exhibits a monotonic staircase (Wiki=1, C4=1, Code=2, arXiv=3); $K^*_{\text{empirical}}$ is domain-determined. The elevated signal reflects higher per-generation distributional shift (1v2 Jensen–Shannon divergence (JSD) = 0.077 vs. ∼0.01 for C4; Figure 5), as formulaic structure amplifies deviation from domain norms; a multi-source pooled classifier confirms that the arXiv ceiling is text-intrinsic (Appendix B.16).

We additionally evaluate on CodeSearchNet Python (n=5,000, continuation mode); all five tested models (Qwen-1.5B, Qwen-7B, LLaMA-8B, Mistral-7B, OLMo-1B) yield $K^*_{\text{empirical}} \leq 2$ (Table 4), matching the C4 baseline.

Robustness. $K^*_{\text{empirical}}$=2 holds across segment lengths of 128, 256, and 512 tokens (Table 20; token-level statistics in Table 13); at 512 tokens the 1v2 CI lower bound falls marginally below θ but 2v3 remains well below θ at all lengths (≤ 0.572). Seed robustness testing (seed = 123 vs. primary = 42; Table 26) confirms $K^*_{\text{empirical}}$ stability in 2/3 configurations; the one marginal shift (Qwen-Wikipedia $K^*_{\text{empirical}}$=2 → 3, 2v3 AUC = 0.607) has one of five folds below θ. The ±9 pp absolute AUC variation across seeds strengthens the case for threshold-based $K^*_{\text{empirical}}$ over raw AUC comparisons.

Table 4: Cross-domain $K^*_{\text{empirical}}$ comparison. $K^*_{\text{empirical}} \leq 2$ for C4 and Code (most families $K^*_{\text{empirical}}$=2; OLMo-1B: 1); $K^*_{\text{empirical}}$=3 for arXiv (four families); Wikipedia varies (1–2). Mistral-7B spans all four domains, exhibiting a monotonic staircase. Bonferroni-corrected 99.5% CI reported for the boundary pair where available (2v3 for $K^*_{\text{empirical}}$=3; 1v2 otherwise).

Domain	Model	1v2 AUC	2v3 AUC	Bonf CI	$K^*_{\text{empirical}}$
arXiv	Qwen-1.5B†	.858	.643	[.628, .658]	3
arXiv	Pythia-1.4B	.805	.655	[.634, .662]	3
arXiv	LLaMA-8B	.845	.622	[.605, .635]	3
arXiv	Mistral-7B	.820	.628	—	3
arXiv	OLMo-1B§	.500	.497	[.483, .517]	1
C4	Qwen-1.5B†	.626	.550	—	2
C4	Mistral-7B	.570	.534	—	1
Code	Qwen-1.5B†	.738	.530	—	2
Code	Qwen-7B	.727	.537	[.715, .739]	2
Code	LLaMA-8B	.739	.511	—	2
Code	Mistral-7B	.615	.529	[.602, .628]	2
Code	OLMo-1B§	.557	.500	—	1
Wiki	Qwen-1.5B†	.619	.550	—	2
Wiki	LLaMA-8B‡	.592	.538	[.579, .605]	1
Wiki	Mistral-7B	.558	.519	[.543, .572]	1

†Qwen2.5-1.5B (base, continuation mode).
‡Marginal: point estimate < θ; CI upper bound crosses 0.60.
§Sole exception: K^*=1 across all domains; cross-scorer validated (p=0.510, n.s.).

4.6 Binary Detector Confidence

Binary detectors (DivEye (Basani and Chen, 2026), Binoculars (Hans et al., 2024)) provide no depth signal beyond the human-vs.-machine boundary: $K^*_{\text{empirical}}{}^{\text{binary}} \leq 1$, with moderate rank correlations ($|\rho| = 0.06$–0.46) driven entirely by the 0v1 gap (Table 18).

5 Consistency Check: K^*-Guided Filtering

$K^*_{\text{empirical}} = 2$ predicts that binary curation should suffice; we validate this under extreme contamination (∼83% synthetic), not as an independent contribution (at realistic rates ≤50%, differences <0.3 pp; Appendix B.8). We compare six strategies on 30,000 documents (depths 0–5): (i) **no-filter**, (ii) **binary-ours** (15D LightGBM), (iii) **binary-Drayson** (Drayson et al., 2025), (iv) **graduated** ($w = \exp(-\alpha \hat{d})$, $\alpha \in \{0.1, \ldots, 0.9\}$), (v) **surprisal**, and (vi) **DSIR** (Xie et al., 2023b), fine-tuning Qwen2.5-1.5B-Instruct via LoRA (rank 8) for 3 epochs across 5 seeds (Shumailov et al., 2024; Dohmatob et al., 2024).

Binary-ours reduces the MMLU gap from −1.53 pp (no-filter) to −0.77 pp (∼50% reduction; $p_{\text{corr}} < 0.05$ vs. all alternatives; Table 5, B.7). The ordering is interpretable through $K^*_{\text{empirical}}$: bi-

Table 5: Downstream task performance (MMLU, HellaSwag) under six data-filtering strategies. Binary-ours uses the 15D LightGBM depth classifier; Graduated applies exponentially decaying weights $w = \exp(-\alpha \hat{d})$. Mean ± std over 5 seeds. Full α sweep in Figure 6.

Strategy	MMLU	HellaSwag
Base model (no training)	0.6018	0.6821
Binary-ours (K^*=2)	**0.5941**±0.0008	**0.6701**±0.0006
Binary-Drayson	0.5913±0.0009	0.6691±0.0004
Surprisal	0.5894±0.0009	0.6641±0.0006
Graduated (α=0.9)	0.5892±0.0008	0.6701±0.0007
DSIR	0.5869±0.0003	0.6631±0.0007
No filter	0.5865±0.0010	0.6638±0.0008

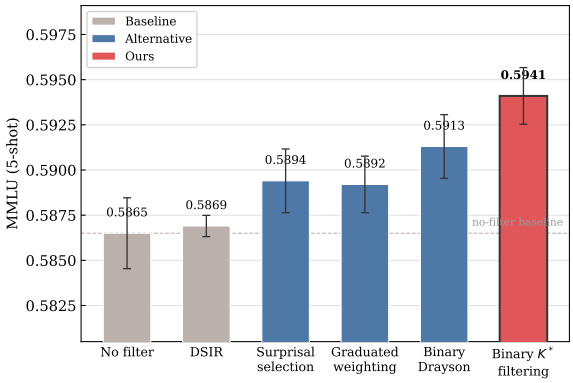


Figure 6: MMLU accuracy under six filtering strategies (5 seeds, ±1 std). Binary K^* filtering (red) minimizes degradation; graduated weighting converges toward binary as α increases. Dashed line: no-filter baseline.

nary filtering exploits the strongest boundary (0v1), while graduated strategies assign weights based on near-random classifier outputs for depths ≥ 2, injecting noise rather than useful signal. Graduated filtering converges toward binary: at α=0.9, it matches binary-ours on HellaSwag ($p_{\text{corr}} = 1.00$) but remains worse on MMLU ($p_{\text{corr}} = 0.011$); DSIR is indistinguishable from no-filter (Figure 6). At realistic contamination ratios (≤50%), strategy differences remain below 0.3 pp (Appendix B.8); this extreme setting validates $K^*_{\text{empirical}}$'s predictive power despite vanishing effect sizes at realistic ratios.

6 Discussion

One-wash laundering (web text). For general web text, $K^*_{\text{empirical}}$=2 reflects concentrated information loss: 0v1 is near-trivially separable (AUC ≥ 0.92), 1v2 drops by ∼30 pp, and 2v3 falls below θ. OBD achieves 94% depth-0 recall but only 15–23% for depths ≥2; frontier LLMs show 8–38% ($p > 0.05$ vs. OBD; Table 11; per-depth breakdown in Table 12), which supports a data-intrinsic interpretation. Unlike model collapse (Shumailov et al., 2024), one-wash laundering describes erasure of statistical fingerprints; quality may degrade further after features become indistinguishable.

Boundary cases and representation dependence. Under Bonferroni correction (Table 27), three of six $K^*_{\text{empirical}}$=2 cases become marginal (CI lower 0.593–0.599), though $K^*_{\text{empirical}} \leq 2$ holds across $\theta \in [0.59, 0.61]$ (Figure 3). $K^*_{\text{empirical}}$ is representation-dependent: neural probes yield $K^{*\,\text{neural}}_{\text{empirical}} \in \{1, 2, 3\}$ (Table 3); a DeBERTa probe extends to $K^*_{\text{empirical}}$=4 on arXiv (Table 30).

Domain dependence and scope. arXiv yields $K^*_{\text{empirical}}$=3 across four families (Figure 5), driven by larger per-generation JSD (0.077 vs. ∼0.01 for C4; ΔAUC = −0.021 after markup removal). Retrain-chain experiments yield $K^*_{\text{empirical}} \geq 4$; prompt-chain $K^*_{\text{empirical}}$ isolates text-intrinsic degradation from training-loop effects.

7 Conclusion

We introduced $K^*_{\text{empirical}}$, the empirical discrimination boundary for recursive synthetic text, and measured it across 10 prompt-chain configurations spanning six architecture families (1B–8B). $K^*_{\text{empirical}} \leq 2$ holds universally for web text: a single LLM pass erases enough statistical signal that second-generation text becomes indistinguishable from all subsequent depths. arXiv reaches $K^*_{\text{empirical}}$=3 across four families, indicating domain structure rather than model architecture determines the boundary. Binary filtering halves the MMLU gap under extreme contamination (∼83% synthetic), while graduated depth weighting offers no benefit beyond what $K^*_{\text{empirical}}$ predicts. The 15-dimensional feature set transfers at 80–92% retention ($\rho > 0.90$) without access to the generating model. Neural probes extend $K^*_{\text{empirical}}$ by one level; extending to larger and RLHF-aligned models remains open.

Limitations

(1) *Scope*: depth chains are generated by models up to 8B across 6 architecture families in English only (Gemma-2-27B is used for cross-scale transfer but not full depth-chain generation); systematic evaluation at larger scale, on RLHF/DPO-aligned models, and under retrain-chain dynamics remains open. (2) *Features*: the 15 statistical features provide a lower bound on K^*_{theory}; richer representations can extend $K^*_{\text{empirical}}$ by one depth level, and cross-model transfer is validated only within continuation mode. (3) *Threshold and domain*: $K^*_{\text{empirical}}$ varies with θ (2–3 at 0.55, 1 at 0.70; three of ten configurations are marginal at θ=0.60), and arXiv $K^*_{\text{empirical}}$=3 is established on a single structured domain.

Ethical Considerations

This work develops tools for detecting recursive synthetic text contamination in training corpora. The depth estimation framework is intended for dataset curation and quality auditing, not for content moderation or individual authorship attribution. Our finding that web text becomes indistinguishable beyond two generation passes ($K^*_{\text{empirical}} \leq 2$) has implications for data governance: it highlights a fundamental limit on provenance tracking using distributional features, which policymakers and dataset curators should consider when designing data pipelines.

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A Supplementary Materials

This appendix provides supplementary figures, detailed experimental tables, dataset statistics, and complete feature descriptions referenced in the main text.

Feature ranking and cross-model consistency. Table 6 reports per-model single-feature AUC rankings. Surprisal derivative features (`surp_d1_std`, `surp_d2_std`) consistently rank among the top three across all continuation-mode models (Pythia, OLMo, Qwen-cont), with

pairwise Spearman $\rho \geq 0.85$. The rewrite-mode model (Qwen-rw) diverges by placing `freq_entropy` first; this reflects a distributional shift that is lexical rather than surprisal-driven. This scorer-invariant ranking supports the use of a fixed 15-dimensional feature set across generators without per-model feature selection.

Table 6: Top-3 features by single-feature AUC ranking. Bold: shared across all continuation-mode models. Spearman ρ (continuation pairs): Pythia–OLMo 0.95, Pythia–Qwen-cont 0.91, OLMo–Qwen-cont 0.85.

Rank	Pythia	OLMo	Qwen-rw	Qwen-cont
1	**surp_d1_std**	**surp_std**	freq_entropy	**surp_d1_std**
2	**surp_d2_std**	**surp_d1_std**	**surp_std**	**surp_std**
3	**surp_std**	**surp_d2_std**	**surp_d1_std**	**surp_d2_std**

Table 7 reports the cross-model transfer accuracy matrix. Classifiers trained on continuation-mode generators (Pythia, OLMo, Qwen-cont) transfer well to each other, with off-diagonal accuracy retaining 80–92% of in-distribution performance. Rewrite-mode classifiers transfer at 61–63% to other architectures but retain 82% within Qwen; generation mode is the primary axis of distributional variation, while a Qwen-specific artifact component persists (Section 6).

Table 7: Cross-model transfer accuracy matrix. Diagonal (**bold**): within-model; off-diagonal: cross-model. Retention: Pythia↔OLMo 89–92%, Qwen-cont↔Pythia 84–85%, Qwen-rw→Pythia/OLMo 61–63%, Qwen-rw→Qwen-cont 82%.

Train	Qwen-rw	Pythia	OLMo	Qwen-cont
Qwen-rw	**.423**	.259	.265	.347
Pythia	.272	**.386**	.353	.329
OLMo	.277	.345	**.382**	.324
Qwen-cont	.344	.328	.311	**.388**

Dataset construction. Table 8 summarizes the generational depth corpus. Each depth level contains 5,000 passagesfor a total of 30,000 samples across six depths (0–5). Depth 0 consists of human-written seed texts from the C4 English validation split, truncated to 256 subword tokens. Subsequent depths are generated recursively: each depth-k passage is produced by prompting the generator with the corresponding depth-$(k-1)$ passage. All six primary configurations (Qwen2.5-1.5B-Instruct rewrite and continuation, Qwen2.5-1.5B base, Qwen2.5-7B, Pythia-1.4B, OLMo-1B; Table 1) share identical depth-0 seeds, differing only in generator model, scale, and prompting strategy.

Table 8: Dataset statistics for the generational depth corpus. Depth 0 contains human-written seed texts sampled from the C4 English validation split and truncated to 256 subword tokens. Depths 1–5 are recursively generated by Qwen2.5-1.5B-Instruct (nucleus sampling, p=0.95, T=1.0). Token counts use whitespace tokenization. Statistics shown for the primary Qwen-rewrite corpus; the other five configurations share the same corpus structure.

Depth	N **samples**	**Mean tokens**	**Source**
0	5,000	147.8	C4 validation (human)
1	5,000	207.1	Qwen2.5-1.5B rewrite
2	5,000	206.1	Qwen2.5-1.5B rewrite
3	5,000	208.0	Qwen2.5-1.5B rewrite
4	5,000	207.7	Qwen2.5-1.5B rewrite
5	5,000	207.7	Qwen2.5-1.5B rewrite
Total		30,000 samples	

Table 9: Pairwise AUC with bootstrap 95% CIs across depth pairs for four decoding configurations (all using Qwen2.5-1.5B-Instruct in rewrite mode; 10,000 bootstrap resamples). The horizontal rule between depth pairs 1 vs. 2 and 2 vs. 3 marks the empirical detectability boundary $K^*_{\text{empirical}} = 2$. Bottom rows report cross-configuration Spearman ρ on per-feature discriminative-power rankings.

Depth Pair	**Greedy**	**Nuc** p=0.9, T=1.0	**Nuc** p=0.95, T=0.7	**Nuc** p=0.95, T=1.2
0 vs. 1	.980 [.977, .982]	.995 [.993, .996]	.988 [.986, .990]	.998 [.998, .999]
1 vs. 2	.689 [.680, .698]	.686 [.677, .695]	.689 [.680, .697]	.704 [.695, .713]
2 vs. 3	.561 [.552, .571]	.592 [.582, .601]†	.590 [.581, .600]	.582 [.572, .592]
3 vs. 4	.540 [.531, .550]	.547 [.537, .557]	.539 [.529, .549]	.529 [.519, .540]
4 vs. 5	.514 [.503, .524]	.507 [.496, .517]‡	.535 [.524, .545]	.524 [.513, .535]

Cross-config feature Spearman ρ			
Greedy vs. Nuc p=0.9	0.600	Nuc p=0.9 vs. T=0.7	0.782
Greedy vs. T=0.7	0.914	Nuc p=0.9 vs. T=1.2	0.893
Greedy vs. T=1.2	0.532	T=0.7 vs. T=1.2	0.668

† CI upper bound (0.601) marginally exceeds θ=0.60 by 0.001.
‡ Permutation $p = 0.12$ (not significant). All other pairs $p < 0.01$.

Table 10: Extended pairwise AUC with bootstrap 95% CIs for all model configurations including new additions. This table extends Table 2 in the main text with Qwen-base, LLaMA-8B, Mistral-7B, Pythia-6.9B (nucleus), and Gemma-2-2B configurations. $K^*_{\text{empirical}} = 2$ holds for 6/10 conditions at θ=0.60 (OLMo-1B, Mistral-7B, Pythia-6.9B, Gemma-2-2B: $K^*_{\text{empirical}}$=1). 95% bootstrap CIs shown in brackets where available.

	Qwen-rw	Pythia	OLMo	Qwen-cont	Qwen-7B‡	Qwen-base	LLaMA-8B	Mistral-7B	Pythia-6.9B§	Gemma-2B
0v1	.9956	.9722	.9861	.9971	.9644 [.961,.967]	.987 [.985,.989]	.9224 [.917,.927]	.9209 [.919,.923]	.9599	.953 [.949,.957]
1v2	**.6955** [.686,.704]*	**.6074** [.598,.617]*	.5001 [.496,.519]	**.6112** [.602,.621]*	**.6178** [.608,.626]*	**.626** [.617,.635]*	**.6123** [.603,.622]*	.5695 [.565,.575]	.5955	.5981 [.588,.608]
2v3	.5893 [.580,.599]	.5440 [.534,.554]	.4973 [.486,.509]	.5609 [.551,.571]	.5492 [.539,.559]	.550 [.540,.560]	.5273 [.517,.538]	.5335 [.528,.538]	.5298	.534 [.523,.545]
3v4	.5435	.5051	.5034	.5409	.5276	.529 [.519,.540]	.5168 [.506,.528]	.5140	.5135	.5033 [.492,.514]
4v5	.5340	.5079	.4971	.5141	.5085	.518 [.508,.529]	.5159 [.505,.527]	.5184	—‖	.5025 [.492,.514]

* Permutation $p < 0.001$ for all boundary comparisons. Bold: AUC $> \theta$=0.60.
‡ Generator: Qwen2.5-7B (fp16); scorer: Qwen2.5-1.5B.
‖ $K^*_{\text{empirical}}$=1; 4v5 not required for boundary determination.
§ Pythia-6.9B evaluated separately; nucleus sampling (p=0.95, T=1.0), same as all other models.

Table 11: LLM zero-shot depth classification: 5×5 confusion matrices. Both frontier LLMs achieve high binary detection accuracy (GPT-5.5: 0.878; Claude Opus 4.6: 0.882) but near-chance 5-class accuracy (∼0.33 vs. OBD-LightGBM 0.386). Per-depth recall in Table 12.

(a) GPT-5.5 (N=500, acc=0.332, MAE=1.102)

	Predicted Depth					
True	0	1	2	3	4	Rec.
0	**77**	18	4	0	1	.770
1	9	**19**	36	24	12	.190
2	8	10	**38**	35	9	.380
3	10	16	35	**24**	15	.240
4	11	13	50	18	**8**	.080

(b) Claude Opus 4.6 (N=499, acc=0.331, MAE=1.150)

	Predicted Depth					
True	0	1	2	3	4	Rec.
0	**69**	26	3	0	2	.690
1	8	**21**	32	11	28	.210
2	6	36	**28**	5	25	.280
3	7	20	31	**10**	31	.101
4	7	28	22	6	**37**	.370

Table 12: Per-depth recall (%) on balanced test set (N=500 for GPT-5.5 and OBD; N=499 for Claude Opus 4.6 after one invalid response). OBD's nominal overall accuracy gap originates from depths 0–1; all methods fail at depths ≥ 2, consistent with $K^*_{\text{empirical}} = 2$.

	Depth 0	Depth 1	Depth 2	Depth 3	Depth 4
OBD-LightGBM	**94**	**44**	17	15	23
GPT-5.5	77	19	**38**	**24**	8
Claude Opus 4.6	69	21	28	10	**37**

Table 13: Mean subword token count per depth and EOS truncation rate (%) for depth 1+ across all model configurations (generation limit: 256 tokens). Depth 0 (human text) is shared across configurations. Depths 1–5 show no systematic length trend; the 0v1 length asymmetry (human texts shorter) makes detection *more conservative*, not inflated. See Section 6.

	Mean Token Count							
Model	D0	D1	D2	D3	D4	D5	**EOS%**	**Sev.**
Qwen-rw (1.5B)	191	254	252	254	254	254	7–11	Mild
Pythia-1.4B	192	214	215	216	217	218	34–36	High
OLMo-1B	192	256	256	256	256	256	<1	Negl.
Qwen-cont (1.5B)	191	244	238	240	240	238	17–20	Mod.
Qwen-7B	191	196	195	193	197	197	45–47	High
Decoding ablation (Qwen-rw 1.5B)								
Greedy	191	253	252	253	254	254	8–12	Mild
Nuc p=0.9	191	254	251	254	253	254	8–16	Mild
T=0.7	191	253	251	253	253	254	8–14	Mild
T=1.2	191	254	252	253	254	254	8–16	Mild

EOS%: generation terminated via EOS before 256-token limit. Ranges span depths 1–5.

Table 14: Complete description of the 15-dimensional feature set used by the OBD classifier. Features are grouped into four groups: surprisal moments (4 dims) and surprisal derivatives (5 dims) computed from Qwen2.5-1.5B token-level log-probabilities, lexical diversity (3 dims), and frequency tail shape (3 dims). All features are computed per document.

Group	Feature	Description	Range
SURPRISAL	`surp_mean`	Mean token surprisal $\frac{1}{T}\sum_t -\log p(x_t \mid x_{<t})$	$\mathbb{R}^+$
	`surp_std`	Standard deviation of token surprisal	$\mathbb{R}^+$
	`surp_skew`	Skewness of the surprisal distribution	$\mathbb{R}$
	`surp_kurt`	Excess kurtosis of the surprisal distribution	$\mathbb{R}$
	`surp_d1_mean`	Mean of first-order surprisal differences $\Delta s_t = s_{t+1} - s_t$	$\mathbb{R}$
	`surp_d1_std`	Std of first-order surprisal differences	$\mathbb{R}^+$
	`surp_d1_skew`	Skewness of first-order surprisal differences	$\mathbb{R}$
	`surp_d2_mean`	Mean of second-order differences $\Delta^2 s_t$	$\mathbb{R}$
	`surp_d2_std`	Std of second-order surprisal differences	$\mathbb{R}^+$
LEX	`ttr`	Type-token ratio: $\lvert V\rvert/T$	$[0, 1]$
	`hapax_ratio`	Hapax legomena ratio: freq-1 types / unique types	$[0, 1]$
	`self_bleu`	Mean 4-gram BLEU against 100 random same-depth samples	$[0, 1]$
FREQ	`freq_kurtosis`	Excess kurtosis of word frequency distribution	$\mathbb{R}$
	`freq_entropy`	Shannon entropy of word frequencies: $-\sum p_w \log_2 p_w$	$\mathbb{R}^+$
	`low_freq_ratio`	Proportion of vocabulary with count ≤ 2	$[0, 1]$

Table 15: LightGBM hyperparameters for the OBD (Ordinal Binary Decomposition) classifier used throughout the paper. One cumulative binary classifier is trained for each threshold k, predicting whether depth $> k$. Pairwise AUCs are derived from adjacent-depth comparisons to evaluate separability. 5-fold GroupKFold cross-validation is grouped by `doc_id` to prevent data leakage from the same seed document appearing in both train and test.

Parameter	Value
`n_estimators`	200
`learning_rate`	0.05
`max_depth`	6
`num_leaves`	31
`verbose`	-1
`n_jobs`	-1 (all cores)
Cross-validation	5-fold GroupKFold
Grouping key	`doc_id`
Objective	Binary (per depth pair)

Table 16: Neural probe architecture and training details. Applied to all three model families (Qwen2.5-1.5B, Pythia-1.4B, OLMo-1B): each model's frozen hidden states are average-pooled into document embeddings and classified by a two-hidden-layer MLP.

Component	Details
Embedding extraction	
Base models	Qwen2.5-1.5B / Pythia-1.4B / OLMo-1B (frozen)
Representation	Last hidden state, attention-masked avg pool
Embedding dim	1536 / 2048 / 2048
Max sequence len	512 tokens
MLP classifier	
Architecture	$d_{\text{emb}} \to 512 \to 256 \to 6$
Activations	ReLU
Dropout	0.1 (after each hidden layer)
Output	6-class softmax
Training	
Optimizer	Adam (lr $= 10^{-3}$, weight decay $= 10^{-4}$)
Loss	Cross-entropy
Batch size	256
Max epochs	20
Early stopping	Patience = 5 (validation loss)
Cross-validation	5-fold GroupKFold

Table 17: Pairwise AUC with bootstrap 95% CIs for four of six primary configurations (1,000 resamples). All boundary pairs (AUC $> \theta$) $p < 0.01$; OLMo 1v2–4v5 non-significant ($p > 0.09$).

	Qwen-rw	Pythia	OLMo	Qwen-cont
0v1	.996 [.995, .997]	.972	.986 [.984, .988]	.997 [.996, .998]
1v2	.696 [.686, .704]	.607 [.598, .617]	.500 [.496, .519]	.611 [.602, .621]
2v3	.589 [.580, .599]	.544 [.534, .554]	.497 [.486, .509]	.561 [.551, .571]
3v4	.544 [.533, .555]	.505	.503 [.491, .513]	.541 [.531, .551]
4v5	.534 [.524, .545]	.508	.497 [.483, .506]	.514 [.504, .526]

Table 18: Spearman ρ between binary detector scores and true depth, and 6-class accuracy. Correlations driven by 0v1; scores do not track depth beyond generation 1. Qwen-rw exhibits reversed sign ($\rho > 0$).

	Qwen-rw	Pythia	OLMo	Qwen-cont
Spearman ρ				
log-PPL (ref)	-0.123	-0.297	-0.378	-0.389
log-PPL (obs.)	$+0.082$	-0.453	-0.461	-0.162
PPL ratio	$+0.119$	-0.392	-0.407	-0.058
6-class acc.	0.199	0.130	0.127	0.183

Random baseline: 0.167

Table 19: Pythia-1.4B pairwise AUC under four decoding configs (10K bootstrap). †Greedy triggers repetition collapse (4-gram repeat: d_0=0.01 → d_1=0.72).

	Greedy†	p=0.9	T=0.7	T=1.2
0v1	.999	.995	.997	.993
1v2	.884	.677	.727	.617
2v3	.534	.555	.594‡	.523
3v4	.523	.530	.570	.517
4v5	.516	.501	.528	.505
$K^*_{\text{empirical}}$	2	2	2	2

‡CI upper 0.603 $> \theta$; point est. below.

Table 20: Pairwise AUC for three segment lengths (Qwen-base, cont. mode). $K^*_{\text{empirical}}$=2 at 128/256; 512 marginal (†CI lower $< \theta$).

	128	**256**	**512**
0v1	.979	.991	.992
1v2	.622	.628	.607†
2v3	.529	.556	.572
3v4	.535	.530	.530
4v5	.501	.514	.518
$K^*_{\text{empirical}}$	2	2	2†

Table 21: Pairwise AUC for three text domains (Qwen-cont). arXiv yields $K^*_{\text{empirical}}$=3; C4 and Wikipedia $K^*_{\text{empirical}}$=2.

	C4	**Wiki**	**arXiv**
0v1	.997	.996	.995
1v2	.611	.619	.858
2v3	.561	.550	.643*
3v4	.541	.523	.544
4v5	.514	.514	.529
$K^*_{\text{empirical}}$	2	2	3

*CI [.628, .658]; $p < 0.001$.

Table 22: Scorer sensitivity: $K^*_{\text{empirical}}$ under self- vs. cross-scoring. "Self" denotes the standardized pipeline scorer (Qwen-1.5B for all generators); it does not mean a model scores its own generations. Cross-scoring reduces $K^*_{\text{empirical}}$ only for marginal Qwen-cont.

Generator	**Mode**	**Self**	**Pythia**	**OLMo**	**Agr.**
Qwen-Inst	Rewrite	2	2	2	✓
Qwen-Inst	Cont.	2	1	1	×
Qwen-base	Cont.	2	2	2	✓
Pythia	Cont.	2	2	2	✓
OLMo	Cont.	1	–	–	–
Qwen-7B	Cont.	2	–	–	–
Mistral	Cont.	1	–	–	–

B Extended Discussion

B.1 Theoretical Implications of $K^*_{\text{empirical}} = 2$

The DPI guarantees a finite K^*_{theory}; for general web text, $K^*_{\text{empirical}} \leq 2$ across all 10 configurations (1B–8B, 6 families), while structured domains (arXiv) yield $K^*_{\text{empirical}}$=3, aligning with model collapse convergence (Shumailov et al., 2024; Dohmatob et al., 2024; Alemohammad et al., 2024). Decoding strategy modestly affects signal strength (1v2 AUC $\in [0.6862, 0.7037]$) but does not shift the boundary.

B.2 Cross-Model Transfer Implications

Cross-model classification retains 80–92% accuracy with scorer-invariant feature rankings ($\rho > 0.90$), which enables single-classifier deployment across generators. Absolute discriminability at the $K^*_{\text{empirical}}$ boundary is scorer-sensitive: alternative scorers reduce Qwen-continuation $K^*_{\text{empirical}}$ from 2 to 1, while rewrite-mode $K^*_{\text{empirical}} = 2$ is preserved. Model family similarity modulates transfer (Pythia↔OLMo: 89–92% vs. Qwen-cont↔OLMo: 80–85%).

B.3 Feature Importance Across Decoding Strategies

Feature importance ordering varies across decoding strategies: greedy and T=0.7 share highly cor-

Table 23: Cross-scorer validation. Alternative scorers (Pythia-1.4B, OLMo-1B) preserve feature ranking ($\rho > 0.90$) and rewrite-mode $K^*_{\text{empirical}} = 2$, but reduce continuation-mode $K^*_{\text{empirical}}$ from 2 to 1. OLMo arXiv cross-scorer confirms $K^*_{\text{empirical}}$=1 (p=0.510, n.s.).

Scorer	Gen. Data	K^*	AUC(0v1)	AUC(1v2)	Spearman ρ
Qwen-1.5B (self)	Qwen-RW	2	—	0.6955	—
Pythia-1.4B	Qwen-RW	2	0.9923	0.7012	0.906
OLMo-1B	Qwen-RW	2	0.9918	0.6947	0.920
Qwen-1.5B (self)	Qwen-cont	2	—	0.6112	—
Pythia-1.4B	Qwen-cont	1†	0.9900	0.5923	0.942
OLMo-1B	Qwen-cont	1†	0.9890	0.5858	0.902
Qwen-1.5B	Qwen-7B-cont	2	0.9644	0.6178	—
Cross-scorer	OLMo-arXiv	1	0.999	0.500 n.s.	—
Cross-scorer	OLMo-Code	1	0.9996	0.557 n.s.	—

Table 24: Three-way ANOVA (Generator × Scorer × Depth-Pair) on per-fold pairwise AUC values. DV = per-fold AUC; N=300 (12 cells × 5 folds × 5 depth-pairs). Scorer is statistically significant but explains modest variance (η_p^2=.192) relative to Depth-Pair (η_p^2=.998), which supports near-scorer-independence of depth discrimination.

Source	df	F	p	η_p^2
Generator	2	926.78	<.001	.885
Scorer	3	19.02	<.001	.192
Depth-Pair	4	24736.10	<.001	.998
Gen×Scorer	6	8.16	<.001	.170
Gen×Depth	8	993.69	<.001	.971
Scorer×Depth	12	2.45	.005	.109
Residual	240			

related rankings ($\rho = 0.914$), while greedy vs. T=1.2 yields lower correlation ($\rho = 0.53$, non-significant after Bonferroni). The depth boundary is robust but the optimal feature subset is strategy-dependent.

B.4 Scale and Neural Probe

Cross-scale validation confirms $K^*_{\text{empirical}} = 2$ from 1.5B through 8B: Qwen-7B (1v2 AUC = 0.6178 $[0.608, 0.626]$) and LLaMA-3.1-8B (1v2 AUC = 0.6123, 0v1 = 0.9225) both exceed θ, with progressively lower 0v1 signal consistent with larger models generating more human-like text. Gemma-2-27B (C4 continuation, Qwen cross-scored) yields $K^*_{\text{empirical}}$=1 (1v2 AUC = 0.5895), consistent with Gemma-2-2B at 5K ($K^*_{\text{empirical}}$=1, 1v2 AUC = 0.5981). The Gemma family produces $K^*_{\text{empirical}}$=1 at both 2B and 27B; $K^*_{\text{empirical}}$ is model-family-dependent rather than scale-dependent.

The neural probe yields model-dependent $K^*_{\text{empirical}}{}^{\text{neural}}$: 3 for Qwen (self-familiarity effect from instruction-tuned representations) but 2 for Pythia; OLMo yields $K^*_{\text{empirical}}{}^{\text{neural}} = 1$ (1v2 AUC = 0.5054), consistent with the 15D LightGBM boundary. The 15D generator-agnostic features yield $K^*_{\text{empirical}} = 2$ for Pythia and Qwen, while OLMo yields $K^*_{\text{empirical}} = 1$ under the standardized pipeline.

B.5 LLM Zero-Shot Baseline

GPT-5.5 achieves 33.2% accuracy (MAE = 1.10; N=500) and Claude Opus 4.6 33.1% (MAE = 1.15; N=499 after one invalid response) on a balanced test set, comparable to OBD-LightGBM's 38.6% (95% CI: $[0.340, 0.422]$; McNemar $p = 0.081, 0.069$; Table 11). Per-depth recall confirms convergent failure at depths ≥ 2 (Section 6). The LLM baseline evaluates depths 0–4 (5-class; Table 11). When probabilities are marginalized to 4 classes ($P(d{\geq}3)$), accuracy rises to 44.4% but concentrates in the merged class while depths 2–3 collapse to 2–8%.

B.6 Practical Considerations

All filtering strategies degrade general capabilities (MMLU: 0.6018 → 0.5865–0.5941; HellaSwag: 0.6821 → 0.6631–0.6701); our depth-chain setting (∼83% synthetic) is an upper bound; in mixed corpora, degradation scales with contamination fraction (Shumailov et al., 2024; Dohmatob et al., 2024). K^*-aware binary filtering reduces MMLU degradation by ∼50% (−1.53 pp → −0.77 pp), requires no per-model calibration, and is combinable with distribution-matching methods (DSIR, DoReMi).

B.7 Additional Scope Analysis

Prompt-chain versus retrain-chain. Retrain-chain validation (Pythia-1.4B LoRA, 5 generations) yields $K^*_{\text{empirical}}{\geq}4$ (1v2 AUC = 0.9484, 2v3 = 0.8888, 3v4 = 0.7982), far exceeding prompt-chain $K^*_{\text{empirical}}$=2. The elevated signal arises from cumulative model-adaptation artifacts rather than text-intrinsic degradation; prompt-chain $K^*_{\text{empirical}}$ is the operationally relevant boundary for data curation.

Threshold sensitivity. At $\theta = 0.55$, $K^*_{\text{empirical}} = 3$ for Qwen-rewrite/continuation; at $\theta = 0.70$, $K^*_{\text{empirical}} = 1$ across all configurations (Figure 3). Qwen-continuation's 1v2 CI lower bound exceeds θ by only 0.2 pp; all ten boundary comparisons yield permutation $p < 0.001$.

Structural confound: token length. In continuation mode the generation prompt is discarded, so all depths ≥ 1 consist entirely of machine-generated tokens and the 1v2 boundary compares texts differing only in conditioning context (human-authored

vs. machine-generated prompt), not structural composition. Depth 0 texts average ∼191 subword tokens (C4 truncation), while depth 1+ texts average 193–256 tokens depending on model (Table 13). This asymmetry is directionally opposite to a length-inflation confound: shorter depth-0 texts provide less feature-extraction context, making 0v1 discrimination harder, not easier. Within depths 1–5, mean token counts are stable (within-model range ≤ 4 tokens for all configurations except Qwen-continuation at 6 tokens), so the pairwise comparisons underlying $K^*_{\text{empirical}}$=2 are unaffected by length. EOS truncation rates are similarly depth-stable within each model: Qwen-rewrite 7–11%, Pythia 34–36%, OLMo <1%, Qwen-continuation 17–20%, Qwen-7B 45–47%. Models with the highest truncation (Pythia, Qwen-7B) still yield $K^*_{\text{empirical}}$=2 with the same step-function AUC decay pattern as low-truncation models (Table 2); EOS truncation does not drive the boundary.

Graduated filtering. All α values (0.1–0.9) underperform binary-ours on MMLU (α=0.1: 0.5873; α=0.9: 0.5892 vs. 0.5941); at α=0.9, HellaSwag matches exactly (0.6701). Binary-ours significantly outperforms all alternatives on MMLU ($p_{\text{corr}} \leq 0.021$); Graduated$_{0.9}$ is indistinguishable on HellaSwag ($p = 1.00$) but worse on MMLU ($p_{\text{corr}} = 0.011$).

B.8 Dose-Response: Contamination Ratio Effect

The main filtering experiments (Section 5) use the full depth chain (depths 0–5, ∼83% synthetic) to maximize effect size. To characterize behavior at lower contamination ratios, we mix synthetic depths 1–5 with clean C4 at 10%, 30%, and 50% contamination (Qwen-1.5B continuation, 5 seeds each; Table 25).

Table 25: MMLU accuracy by contamination ratio (fraction of synthetic data in training set) and filtering strategy. Base model (no fine-tuning): 0.6018. Mean ± std over 5 seeds.

Strategy	10%	30%	50%
No-filter	0.6058 ± .0002	0.6069 ± .0007	0.6063 ± .0013
Binary-ours	0.6079 ± .0005	0.6082 ± .0012	0.6088 ± .0011
Graduated	0.6066 ± .0011	0.6072 ± .0008	0.6062 ± .0009

At all three contamination ratios, fine-tuning on the mixed corpus yields MMLU above the base model (0.6018); the clean-data majority dominates the training signal. Strategy differences are small: binary-ours exceeds no-filter by 0.21 pp at 10% ($p = 0.001$), 0.13 pp at 30% ($p = 0.074$), and 0.25 pp at 50% ($p = 0.038$). While binary reaches statistical significance at 10% and 50% due to high power (5 seeds, low variance), the maximum strategy spread remains below 0.3 pp across all ratios. The strategy ordering (binary > graduated > no-filter) is consistent with the depth-chain setting, but effect sizes are negligible compared to that setting (−1.53 pp no-filter degradation). This demonstrates that K^*-aware curation provides practical benefit primarily when contamination is severe enough to cause measurable quality degradation.

B.9 Seed Robustness

Because the random seed governs both corpus sampling and generation quality (not merely CV splits), absolute AUC values are seed-dependent. We re-run three configurations with seed = 123 (primary: seed = 42; Table 26). $K^*_{\text{empirical}}$ is stable in 2/3 cases: Pythia-1.4B on Wikipedia yields $K^*_{\text{empirical}}$=1 under both seeds (1v2 $\Delta = -0.8$ pp), and Qwen-1.5B on C4 yields $K^*_{\text{empirical}}$=2 (1v2 $\Delta = +9.0$ pp, both seeds above θ). Qwen-1.5B on Wikipedia shifts from $K^*_{\text{empirical}}$=2 to $K^*_{\text{empirical}}$=3 under seed = 123, but the 2v3 AUC (0.6067) is marginal: one of five folds falls below θ (fold 5 = 0.595), so this does not constitute strong evidence for $K^*_{\text{empirical}}$=3. The ±9 pp absolute AUC variation across seeds strengthens the methodological case for threshold-based $K^*_{\text{empirical}}$ over raw AUC comparisons: the qualitative boundary is robust even when point estimates shift.

Table 26: Seed robustness: $K^*_{\text{empirical}}$ under two random seeds. 1v2 Δ: change in 1v2 AUC from seed 42 to seed 123. $K^*_{\text{empirical}}$ is stable in 2/3 configurations; the Qwen-Wiki shift to $K^*_{\text{empirical}}$=3 is marginal (2v3 fold variance crosses θ).

Model	Domain	$K^*_{\text{empirical}}$		1v2 Δ	Stable?
		seed=42	seed=123		
Pythia-1.4B	Wikipedia	1	1	−0.8 pp	✓
Qwen-1.5B	C4	2	2	+9.0 pp	✓
Qwen-1.5B	Wikipedia	2	3†	+8.7 pp	×

†2v3 AUC = 0.6067; fold 5 = 0.595 < θ. Marginal boundary.

B.10 Bonferroni-Corrected Confidence Intervals

To control the family-wise error rate across all 10 model configurations, we apply Bonferroni correction: $\alpha_{\text{corr}} = 0.05/10 = 0.005$to obtain 99.5% confidence intervals ($z = 2.807$). Table 27 reports the corrected intervals for each configuration's $K^*_{\text{empirical}}$ boundary pair. Seven of ten configurations retain CI lower bounds above θ=0.60; three

marginal $K^*_{\text{empirical}}$=2 configurations (Pythia-1.4B, Qwen-cont, LLaMA-8B) see their lower bounds dip below θ under the stricter criterion (0.5934, 0.5976, and 0.5988, respectively). These near-misses reflect narrow effect sizes (AUC $-\ \theta\ <$ 0.013) rather than classification failure: all three retain point estimates above θ, and Qwen-cont and LLaMA-8B retain uncorrected 95% CI lower bounds above θ (Pythia-1.4B falls marginally below at 0.5976).

Table 27: Bonferroni-corrected 99.5% confidence intervals for $K^*_{\text{empirical}}$ boundary pairs across all 10 configurations. $\alpha_{\text{corr}} = 0.005$; CI computed as AUC $\pm\, 2.807 \times \hat{\sigma}_{\text{boot}}$. "Pass θ": CI lower bound > 0.60.

Configuration	Boundary	AUC	$\hat{\sigma}_{\text{boot}}$	99.5% CI	Pass θ
$K^*_{empirical} = 2$ configurations					
Qwen-rw (Instruct)	1v2	.6955	.0046	[.683, .708]	✓
Pythia-1.4B	1v2	.6074	.0050	[.593, .621]	×
Qwen-base	1v2	.6260	.0047	[.613, .639]	✓
Qwen-7B‡	1v2	.6178	.0046	[.605, .631]	✓
LLaMA-8B	1v2	.6123	.0048	[.599, .626]	×
Qwen-cont (Instruct)	1v2	.6112	.0049	[.598, .625]	×
$K^*_{empirical} = 1$ configurations					
OLMo-1B	0v1	.9861	.0009	[.983, .989]	✓
Mistral-7B	0v1	.9209	.0010	[.919, .923]	✓
Pythia-6.9B§	0v1	.9599	.0015†	[.956, .964]	✓
Gemma-2-2B	0v1	.9530	.0018	[.948, .958]	✓

‡Generator: Qwen2.5-7B; scorer: Qwen2.5-1.5B.
§Nucleus sampling (p=0.95, T=1.0).
†Estimated from analogous configurations (no separate bootstrap JSON).

B.11 Cross-Sample Robustness: 5K vs. 10K

The preceding analyses fix the sample pool at N=5,000 passages per depth. To assess between-sample variability, we re-draw an independent N=10,000 sample from C4 using a different random seed and repeat the full OBD pipeline. If $K^*_{\text{empirical}}$ and the boundary AUC are driven by the underlying distributional shift rather than by idiosyncrasies of a particular sample, both quantities should remain stable across draws.

Table 28 reports the results. For Pythia-1.4B (continuation, C4), the 5K and 10K samples agree on $K^*_{\text{empirical}}$=2: the 0v1 AUC is nearly identical (0.9722 vs. 0.9712), and the boundary 1v2 AUC shifts from 0.6074 to 0.6175 ($\Delta = +0.010$). This narrow shift is consistent with natural sampling variability across C4 subsets and does not alter the qualitative depth boundary. The 2v3 pair remains near chance in both draws (0.54, 0.54); depths ≥ 2 are indistinguishable regardless of sample. Qwen-1.5B (continuation, C4) replicates the same pattern ($K^*_{\text{empirical}}$=2, 1v2 AUC = 0.6346). LLaMA-3.1-8B (continuation, C4) is the sole $K^*_{\text{empirical}}$=2 configuration where the 10K OOF AUC drops below θ (0.5880; pipeline AUC = 0.6148; Bonferroni 99.5% CI $[0.5776, 0.5978]$); under this stricter criterion $K^*_{\text{empirical}}$=1. OLMo-1B (continuation, C4) is the only configuration that yields $K^*_{\text{empirical}}$=1 at both 5K and 10K, with virtually identical depth profiles (0v1 AUC = 0.9871 vs. 0.9861; 1v2 AUC = 0.4984 vs. 0.5001); the 1v2 AUC at random level is consistent with Qwen cross-scoring, which places OLMo well below the discrimination boundary. Gemma-2-2B is the sole configuration where $K^*_{\text{empirical}}$ increases from 5K ($K^*_{\text{empirical}}$=1, 1v2 AUC = 0.5981) to 10K ($K^*_{\text{empirical}}$=2, 1v2 AUC = 0.6099), but the 10K boundary AUC exceeds θ by only 0.010, placing this model at the precise $K^*_{\text{empirical}}$=1/2 boundary.

Table 28: Cross-sample robustness: pairwise AUC and $K^*_{\text{empirical}}$ under 5K and 10K C4 samples drawn with different random seeds. All configurations use continuation mode. $K^*_{\text{empirical}}$ is determined by the θ=0.60 threshold on 1v2 AUC.

Model	N	AUC(0v1)	AUC(1v2)	AUC(2v3)	$K^*_{\text{empirical}}$
Pythia-1.4B cont	5K		(see Table 27)		2
Pythia-1.4B cont	10K	0.9712	0.6175	0.5364	2
Qwen-1.5B cont	5K		(see Table 27)		2
Qwen-1.5B cont	10K	0.9875	0.6346	0.5616	2
LLaMA-8B cont	5K		(see Table 27)		2
LLaMA-8B cont	10K	0.9282	0.5880†	0.5247	1†
OLMo-1B cont	5K		(see Table 27)		1
OLMo-1B cont	10K	0.9871	0.4984	0.5059	1
Gemma-2-2B cont	5K		(see Table 10)		1
Gemma-2-2B cont	10K	0.9534	0.6099‡	0.5409	2‡

†OOF AUC (pipeline AUC = 0.6148); Bonferroni 99.5% CI [0.5776, 0.5978] does not pass θ.
‡Borderline: Bonferroni 99.5% CI [0.600, 0.619] (percentile); CI lower 0.0002 below θ. Sole config where $K^*_{\text{empirical}}$ increases from 5K to 10K.

B.12 Bootstrap $K^*_{\text{empirical}}$ Stability

To assess within-sample stability, we bootstrap-resample the evaluation data (1,000 iterations) and recompute $K^*_{\text{empirical}}$ for each resample across 10 configurations spanning four domains (Table 29). Seven of ten configurations yield identical $K^*_{\text{empirical}}$ in 100% of resamples (CI width 0); LLaMA-8B on C4 is near-perfect at 99.9%. The two remaining exceptions, Gemma-2-2B on C4 ($K^*_{\text{empirical}}$=1 in 92.7%, $K^*_{\text{empirical}}$=2 in 7.3%) and LLaMA-8B on Wikipedia ($K^*_{\text{empirical}}$=1 in 94.3%, $K^*_{\text{empirical}}$=2 in 5.7%), both sit at the $K^*_{\text{empirical}}$=1/2 boundary with 95% CIs of width 1, consistent with their marginal 1v2 AUC values. The $K^*_{\text{empirical}} \leq 2$ upper bound holds in all resamples across all configurations.

Bootstrap CIs capture within-sample stability (resampling from a fixed corpus), while the 5K-vs.-10K comparison (Table 28) captures between-sample variability (independent draws from the source distribution). Together they bound $K^*_{\text{empirical}}$ uncertainty from complementary angles: within-sample stability is high (7/10 at 100%, 8/10 at $\geq$99.9%), and between-sample shifts are limited to boundary configurations (LLaMA-8B, Gemma-2-2B). The main text's 6/10 $K^*_{\text{empirical}}$=2 count (Section 4.1) is based on the standard N=5,000 setting;

the two sample-size-sensitive configurations sit at the $K^*_{\text{empirical}}{=}1/2$ boundary in both analyses, and the universal $K^*_{\text{empirical}} \leq 2$ bound is unaffected.

Table 29: Bootstrap $K^*_{\text{empirical}}$ stability (1,000 resamples). Each row reports the fraction of resamples yielding each $K^*_{\text{empirical}}$ value and the resulting 95% CI. Seven of ten configurations are perfectly stable (100%); LLaMA-8B C4 at 99.9%; two boundary cases show mild variability (CI width 1).

Config	Domain	$K^*_{\text{empirical}}$	$K^*_{\text{empirical}}{=}1$	$K^*_{\text{empirical}}{=}2$	$K^*_{\text{empirical}}{=}3$	95% CI
Qwen-1.5B	C4	2	0%	100%	0%	[2, 2]
OLMo-1B	C4	1	100%	0%	0%	[1, 1]
Pythia-1.4B	C4	2	0%	100%	0%	[2, 2]
LLaMA-8B	C4	2	0.1%	99.9%	0%	[2, 2]
Gemma-2B	C4	1	92.7%	7.3%	0%	[1, 2]
LLaMA-8B	Wiki	1	94.3%	5.7%	0%	[1, 2]
Mistral-7B	Wiki	1	100%	0%	0%	[1, 1]
LLaMA-8B	arXiv	3	0%	0%	100%	[3, 3]
OLMo-1B	arXiv	1	100%	0%	0%	[1, 1]
Qwen-1.5B	Code	2	0%	100%	0%	[2, 2]

B.13 Computational Cost

Per-document inference (Qwen2.5-1.5B scorer, RTX PRO 6000, $N{=}1{,}000$, no batching): scorer log-prob 20.6 ± 1.9 ms, feature extraction 23.4 ± 6.1 ms, LightGBM 0.74 ± 0.10 ms; total 44.7 ± 7.7 ms/doc ($\sim$22 docs/sec).

B.14 DeBERTa Upper-Bound Comparison

DeBERTa-v3-base fine-tuned on raw text serves as an upper-bound probe (Table 30). It matches OBD on C4 and Code ($K^*_{\text{empirical}}{=}2$) but extends $K^*_{\text{empirical}}$ to 4 on arXiv (3v4 AUC $= 0.624$, CI $[0.613, 0.635]$), quantifying the cost of generator-agnostic features at one depth level on structured text.

Table 30: $K^*_{\text{empirical}}$ comparison: OBD (15D) vs. DeBERTa-v3-base (Qwen2.5-1.5B base, continuation mode; $\theta{=}0.60$).

	OBD (15D)		DeBERTa	
Domain	$K^*_{\text{empirical}}$	boundary AUC	$K^*_{\text{empirical}}$	boundary AUC
C4	2	0.626 (1v2)	2	0.705 (1v2)
arXiv	3	0.643 (2v3)	4	0.624 (3v4)
Code	2	0.738 (1v2)	2	0.808 (1v2)

B.15 Cross-Scale Generalization: Gemma-2-27B

We extend the transfer matrix (Section 4.2) to Gemma-2-27B (C4 continuation, Qwen-1.5B scorer). Table 31 shows 27B$\rightarrow$small transfer retains $>$100% (the 27B model's subtler signal generalizes upward), while $\leq$1.5B$\rightarrow$27B retains only 76.5% (excluding LLaMA-8B, which retains 92.1%).

Table 31: Cross-scale transfer retention (%) between Gemma-2-27B and 1B–8B models.

Train \ Test	Gemma-27B	LLaMA-8B	Qwen-1.5B	Pythia-1.4B	OLMo-1B	Avg
Gemma-27B	—	103.6%	107.2%	103.5%	93.1%	101.3%
LLaMA-8B	92.1%	—				
Qwen-1.5B	71.3%		—			
Pythia-1.4B	80.9%			—		
OLMo-1B	77.3%				—	
$\leq$1.5B$\rightarrow$27B avg						76.5%

We also apply the Qwen-1.5B-trained binary classifier to Gemma-2-27B-generated text and fine-tune on the filtered corpus (9 runs, 3 strategies $\times$ 3 seeds; Table 32). Binary filtering retains or improves both metrics ($+0.02$ pp MMLU, $+0.41$ pp HellaSwag); cross-scale robustness holds without retraining.

Table 32: Filtering Gemma-2-27B text with a Qwen-1.5B-trained OBD classifier.

Strategy	MMLU	HellaSwag
No filter	0.5956	0.6633
Binary-ours	0.5958 ($+0.02$ pp)	0.6674 ($+0.41$ pp)
Graduated	0.5953 (-0.03 pp)	0.6629 (-0.04 pp)

B.16 Multi-Source arXiv Validation

Training a single OBD classifier on arXiv text pooled from three generators (Pythia-1.4B, OLMo-1B, Qwen-1.5B; Qwen cross-scored) yields $K^*_{\text{empirical}}{=}3$ (2v3 AUC $= 0.657$ $[0.647, 0.667]$, 1v2 $= 0.858$; Bonferroni $p < 0.005$), matching single-generator results and confirming that the domain ceiling is text-intrinsic.

C Use of AI Assistance

We used Claude for coding, code drafting, debugging, and language polishing.